Proposition 12: cut distance and spectral distance bound each other

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Proposition12_CutVsSpectral`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Proposition 12 (Section 3.4, p. 11, l. 385–386): *for all graphs G, H of the same order,* $\delta_\square(G,H) \le \delta_2(G,H) \le 2\sqrt{\delta_\square(G,H)}$. "The second inequality uses a result of Nikiforov [Nik09]" (l. 387). The proof is at the beginning of Appendix F (pp. 44–45, l. 1382–1397). It shows, for all $A \in \mathbb{R}^{V\times W}$ with $m = |V|$, $n = |W|$ (p. 44, l. 1384),

$$\tfrac{1}{\sqrt{mn}}\|A\|_\square \le \|A\|_2 \le 2\sqrt{\|A\|_{(\infty)}\|A\|_\square}, \qquad \|A\|_{(\infty)} := \max_{v\in V, w\in W} |A_{vw}| \text{ (l. 1385)},$$

the first by Cauchy–Schwarz and the second by citing Nikiforov [Nik09, Theorem 3], and applies it to $A^\pi_G - A_H$, whose entries lie in $[-1,1]$, with the normalisations $\delta_\square = n^{-2}d_\square$, $\delta_2 = n^{-1}d_2$ (l. 1385–1387).

Lean. `GsimOT.proposition12`: for finite types V, W with $|V| = |W|$ and simple graphs G, H,

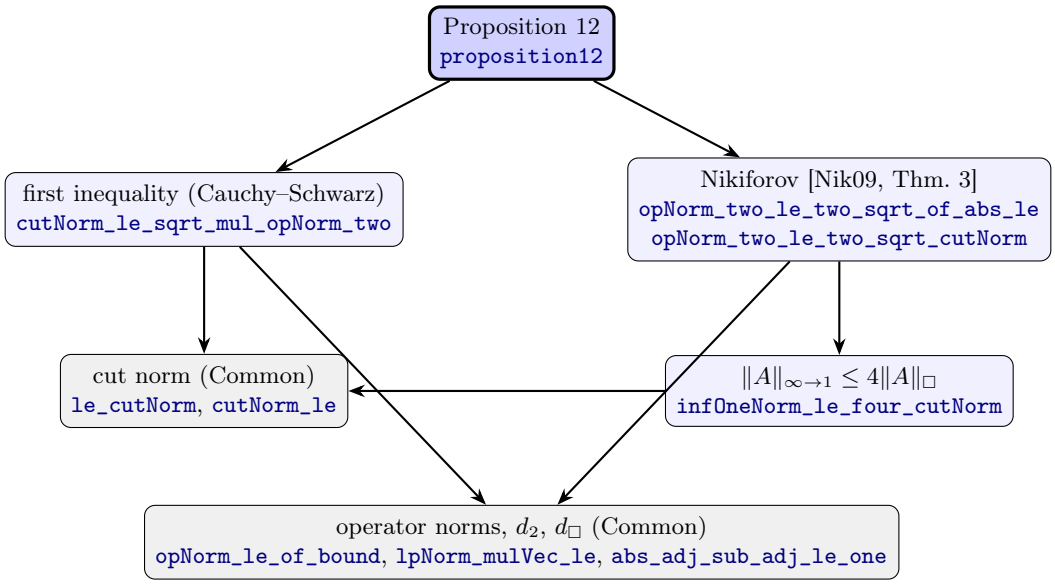
$$\delta_\square(G,H) \le \delta_2(G,H) \quad \text{and} \quad \delta_2(G,H) \le 2\sqrt{\delta_\square(G,H)},$$

with $\delta_\square$ = `deltaCut G H` and δ_2 = `deltaPQ 2 2 G H`. This is exactly the statement. The matrix inequalities are `cutNorm_le_sqrt_mul_opNorm_two` ($\|A\|_\square \le \sqrt{mn}\,\|A\|_2$) and `opNorm_two_le_two_sqrt_cutNorm` (Nikiforov's inequality, *proved* here).

2 Where the proof lives

File	Content
`Proposition12.lean`	the entrywise maximum norm (`maxEntryNorm`), the two matrix inequalities, $\|A\|_{\infty\to 1} \le 4\|A\|_\square$ (`infOneNorm_le_four_cutNorm`), and `proposition12`.
`Common/CutNorm.lean`	the cut norm `cutNorm` ($\max_{S,T} \lvert\sum_{s\in S, t\in T} A(s,t)\rvert$, `le_cutNorm`, `cutNorm_le`), $d_\square$, $\delta_\square$ (`dCut`, `deltaCut`).
`Common/OperatorNorm.lean`, `Common/AlignedDistances.lean`	$\|\cdot\|_{2\to 2}$ (`opNorm`, `opNorm_le_of_bound`, `lpNorm_mulVec_le`), $A^\pi_G - A_H$ (`alignedDiff`), d_2, δ_2 (`distPQ`, `deltaPQ`).

3 Proof layout



Arrows mean "uses". Blue: this folder; grey: `GsimOT/Common`. Nothing is cited: Nikiforov's inequality, which the paper cites, is proved.

4 The proof step by step

First inequality (pp. 44–45, l. 1388–1396). $\|A\|_\square = \max_{S,T} |\langle \mathbf{1}_S, A\mathbf{1}_T\rangle| \le \max \|\mathbf{1}_S\|_2 \|A\mathbf{1}_T\|_2 \le \sqrt{mn}\,\|A\|_2$, with $\|\mathbf{1}_U\|_2 \le \sqrt{|U|}$. `cutNorm_le_sqrt_mul_opNorm_two` follows this: the indicator vector `ind`, $\|A\mathbf{1}_T\|_2 \le \|A\|_2\sqrt{n}$ (`lpNorm_mulVec_le`), and Cauchy–Schwarz `sq_sum_le_card_mul_sum_sq` for the sum over S.

Second inequality (p. 45, l. 1397, cited). The paper cites [Nik09, Theorem 3]. The Lean proves the version with an arbitrary entrywise bound M, `opNorm_two_le_two_sqrt_of_abs_le`: by the Schur test, $\|Ax\|_2^2 = x^\top (A^\top A)x \le \max_j \sum_k |(A^\top A)_{jk}| \cdot \|x\|_2^2$, and $\sum_k |(A^\top A)_{jk}| = \sum_i A_{ij}(As)_i \le M\|A\|_{\infty\to 1} \le 4M\|A\|_\square$ for a sign vector s (`infOneNorm_le_four_cutNorm`: split sign vectors into positive and negative parts, giving four cut values). `opNorm_two_le_two_sqrt_cutNorm` is the case $M = \|A\|_{(\infty)}$.

Application to graphs (p. 44, l. 1385–1387). In `proposition12`: for the first inequality take a bijection π minimising $\|A^\pi_G - A_H\|_2$; $d_\square \le \|A^\pi_G - A_H\|_\square \le n\, d_2$, i.e. $\delta_\square = d_\square/n^2 \le d_2/n = \delta_2$. For the second take π minimising the cut norm; $M = 1$ by `abs_adj_sub_adj_le_one`, so $d_2 \le 2\sqrt{d_\square}$, i.e. $\delta_2 \le 2\sqrt{\delta_\square}$. Minima over the finite set of bijections exist (`exists_eq_ciInf_of_finite`); $n = 0$ is trivial.

5 Assumptions and trusted base

No cited results; the hypothesis is $|V| = |W|$. Axioms: `propext`, `Classical.choice`, `Quot.sound`.

6 Notes on the formalisation

- Nikiforov's inequality [Nik09, Thm. 3] is proved instead of cited, in the more general form with an arbitrary entrywise bound M (`opNorm_two_le_two_sqrt_of_abs_le`); the application to graphs uses $M = 1$.
- The first matrix inequality is stated as $\|A\|_\square \le \sqrt{mn}\,\|A\|_2$ (`cutNorm_le_sqrt_mul_opNorm_two`), the paper's $\frac{1}{\sqrt{mn}}\|A\|_\square \le \|A\|_2$ multiplied by $\sqrt{mn}$.

7 How to check this result

```
lake build GsimOT.Proposition12_CutVsSpectral.Proposition12
lake env lean scripts/CheckAxioms.lean   # prints the axioms of GsimOT.proposition12
```

The table in Proposition12_correspondence.pdf lists every step with file and line.